



JECRCTM
UNIVERSITY
BUILD YOUR WORLD



JECRCTM
UNIVERSITY

BUILD YOUR WORLD

Software Engineering & Project Management

Lecture Notes

Meghna Sharma

CSE



JECRC[™]
UNIVERSITY

BUILD YOUR WORLD

Topics:-

- ❑ Introduction of Software Engineering
- ❑ What is System?
- ❑ Software Characteristics
- ❑ Software Crisis
- ❑ Job Responsibilities of Programmer & Engineers
- ❑ SDLC

SOFTWARE ENGINEERING

Is the application of the systematic, disciplined and quantifiable approach to the development, operation and the maintenance of the software? It encompasses techniques & procedures often regulated by the

system development process with the purpose of improving reliability & the care.

Need of Software Engineering

- Large software
- Scalability
- Cost

- Dynamic Nature
- Quality Management

Characteristics of good software

A software product can be judged by what it offers and how well it can be used. This software must satisfy on the following grounds:

- **Operational:-** Budget, Usability, Efficiency, correctness, functionality, dependability, safety & security.



- **Transitional:-** Portability, interoperability, reusability™, adaptability.
- **Maintenance:-** Modularity, Maintainability, Flexibility, Scalability.



JECRC[™]
UNIVERSITY

BUILD YOUR WORLD

SYSTEM ENGINEERING

- **Systems engineering** is an interdisciplinary field of **engineering** and **engineering** management that focuses on how to design, integrate, and manage complex **systems** over their life cycles. ... **Systems engineering** deals with work processes, optimization methods, and risk management tools in such projects.

- Characteristics: - 1. Synergism
2. Adaptability
3. Compromise

SOFTWARE CHARACTERISTICS

- 1. Software is developed or engineered .It is not manufactured in the classical sense.

Although some similarities exist b/w software development & h/w development but the two activities r different.

2. Quality comes in terms of hardware and quality occurs in terms of software.

SOFTWARE CHARACTERISTICS

- Software does not wear out but it does deteriorate.
- The relationship for software is called as “bathtub curve”
- It indicates that h/w curve is spike curve.

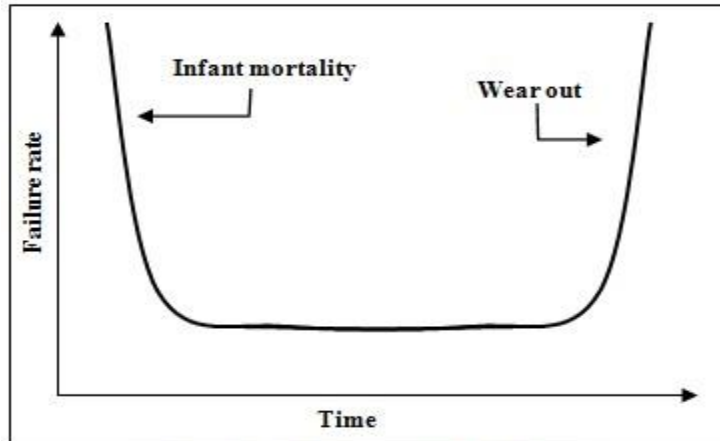


Figure 01: Failure curve for hardware

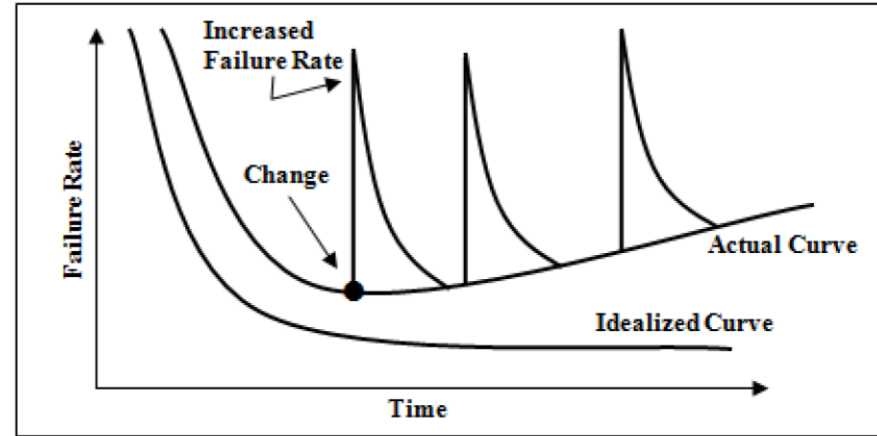


Figure 02: Failure Curves for software

SOFTWARE CHARACTERISTICS

- Although the industry is moving towards component based construction ,most of the s/w continues to be custom built.

- A software component should be designed & implemented so that it can be reused in many different programs.

SOFTWARE CRISIS

- A set of problems encountered in the development of a computer software.
- It is inability to develop a software on budget & within requirement .
- Due to the lack of communication between the s/w & h/w.

- Increase in the cost of software as compared to the H/w.
- Lack of understanding of the problem & the environment .
- Cost & schedule are the major cause of S/w crisis.

Cont...

- Software crisis explained as a mismatch in software deliverables and the expectations a user have from the computer system in terms of its capacity to

work. Due to the software crisis, programmers had to struggle always in order to keep pace.

- The problems attached with low quality software like inefficient software development, improper functioning of the software systems along with the unhappiness within the software users were the major causes for the software crisis

JOB RESPONSIBILITIES OF A SOFTWARE DEVELOPER

- Coordinate with the Technical Director on current programming tasks.
- Collaborate with other programmers to design and implement features.
- Quickly produce well-organized, optimized, and documented source code.
- Create and document software tools required by artists or other developers.
- Debug existing source code and polish feature sets.
- Contribute to technical design documentation.
- Work independently when required.



IECRC
UNIVERSITY

BUILD YOUR WORLD

- Continuously learn and improve skills.
- Attention to detail is essential and all tasks must be carried out to the highest standard.

SDLC (Software Development Life Cycle)

SDLC stands for **Software development life cycle**. It is a process that describes how to develop, design and maintain the software project ensuring that all the functional & user requirement, goals and objective are met. This methodology improves the quality of the software project and over all process of software development.



Figure:- SDLC
Phases

- 1 .Requirement Analysis :

Explicit & implicit s/w requirement specification obtained by studying problem

domain, operational timelines and procedure environment.

SRS is prepared in this phase only



2 .DESIGN SPECIFICATION

- Software development plan, implicit and explicit Software requirement and the software architecture.
- INTERNAL DESIGN: Algorithm& pseudo code
- EXTERNAL DESIGN: User interface/w interface, global and control structure.

3 . DEVELOPMENT

- This is the longest phase of SDLC as in this phase of SDLC actual project is developed and build. This phase of SDLC includes:-
- Actual code gets written.
- Demonstration of work accomplished to Business analyst for future modification as per requirement.
- Unit testing is performed i.e. verification of code as per requirements.

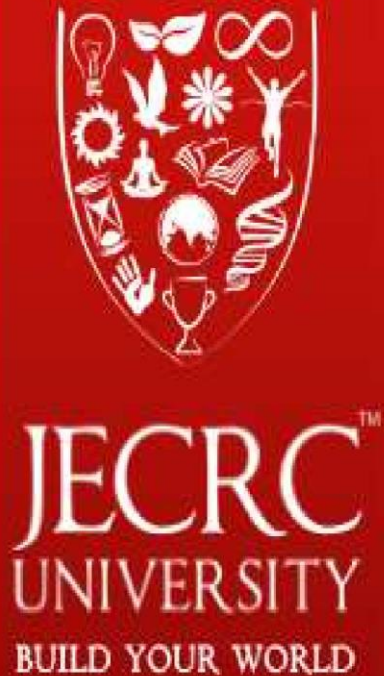
TESTING

- The testing strategy is involved in almost all stages of SDLC. However this phase of SDLC refers to the only testing of system where bugs/defect of the system are reported, tracked and fixed. The system/project is migrated to a test environment and different type of testing is performed like functional, integration, system and acceptance. This is performed until the project reaches the quality standards as specified in SRS. This phase of SDLC includes:
 - System is tested as a whole
- Different type of testing to report & fix bugs.



DEPLOYMENT & MAINTENANCE

- In this phase of SDLC, once the system is tested, it is ready to go live. The system may be first released for limited users, and tested in a real business environment for UAT (user acceptance testing). This phase of SDLC includes System is ready to be delivered.
- The system is installed and put into use.
- Correction of errors that were not caught before.



- Improving system and ultimately enhanced system in data center.



Thank You!